← Back **Graded quiz - Using probability distributions**
Graded Assignment · 30 min

**Due** Mar 20, 11:59 PM CET

**1.** For your Lean Six Sigma project you collect data on 50 calls in a call center.

What describes the population the best?

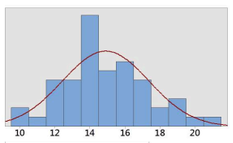
1 point

- All incoming calls.
- All call centers.
- All potential customers.
- All call center agents.

**2.** Consider the provided output that gives the patients Length of Stay.

Which statement is **false**?

1 point




| | |
|---|---|
| Mean | 14.900 |
| StDev | 2.460 |
| Variance | 6.051 |
| Skewness | 0.317061 |
| Kurtosis | -0.035204 |
| N | 50 |
| Minimum | 10.000 |
| 1st Quartile | 13.000 |
| Median | 14.500 |
| 3rd Quartile | 16.250 |
| Maximum | 21.000 |
| 95% Confidence Interval for Mean | |
| 14.201 | 15.599 |
| 95% Confidence Interval for Median | |
| 14.000 | 16.000 |
| 95% Confidence Interval for StDev | |
| 2.055 | 3.065 |

- The interquartile range of Length of Stay equals 3.25.
- 95% of the data points lie within 14.201 and 15.599.
- There is a 95% probability that the population mean lies within 14.201 and 15.599.
- The Length of Stay was collected for 50 patients.

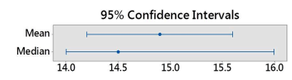
**3.** Which measure **cannot** be used to quantify dispersion?

1 point

- Standard deviation.
- Range.
- Interquartile range.
- Third quartile.

**4.** What can be concluded based on this confidence interval?

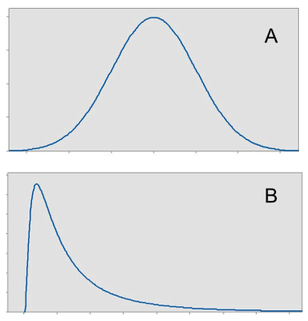
1 point




- The mean is smaller than the median as its interval is more narrow.
- There is uncertainty around the estimates of the mean and median.
- 95% of the data lie within these intervals.
- The estimates of the mean and median are equal.

**5.** Which graph(s) show a skewed distribution?

1 point

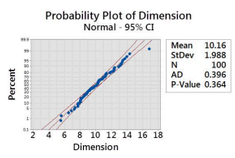



- Neither A nor B.
- Both A and B.
- Only A.
- Only B.

**6.** You have measured your CTQ 'dimension' and made this probability plot.

Which distribution does the data follow?

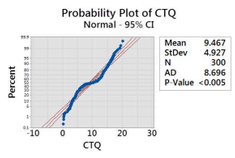
1 point




- Weibull.
- Lognormal.
- Normal.
- Cannot be determined.

**7.** What is a likely explanation for the pattern in this probability plot?

1 point

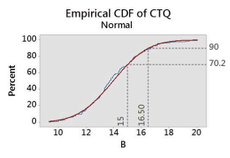



- Bimodality.
- Weibull distribution.
- Skewness.
- Outliers.

**8.** You measured your CTQ.

What can be concluded based on this ECDF (empirical cumulative distribution function)?

1 point




- 15 percent of the population is smaller than 70.2.
- 19.8 percent of the population lie between 15 and 16.50.
- 70.2 percent of the population is larger than 15.
- 90 percent of the population is exactly equal to 16.50.

**9.** You measure the dimension of a product. It is known that dimension follows a normal distribution with mean 10 millimetres, and a standard deviation of 1 millimetre.

Suppose we measured the dimension of 100 products.

Which of the following statements is **false**?

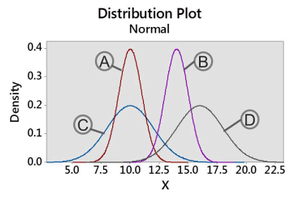
1 point

- Of the 100 products, around 50 will be smaller than 10.
- Around 95 products will have a dimension between 8 and 12.
- Around 99 products will have a dimension between 6 and 14.
- Around 68 products will have a dimension between 9 and 11.

**10.** The probability distribution of 4 datasets (A,B,C and D) are displayed.

Which of these distributions has the **lowest** location parameter and the **lowest** spread parameter?

1 point




- A
- B
- C
- D